VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY

Deep Learning and Its Applications

Food-11 Image Classification

CO3133 · Assignment II
Supervisor: LÊ THÀNH SÁCH

Team members
HOÀNG XUÂN BÁCH · 2352082
NGUYỄN VIỆT HÙNG · 2352424
LÊ NGUYỄN GIA PHÚC · 2352931

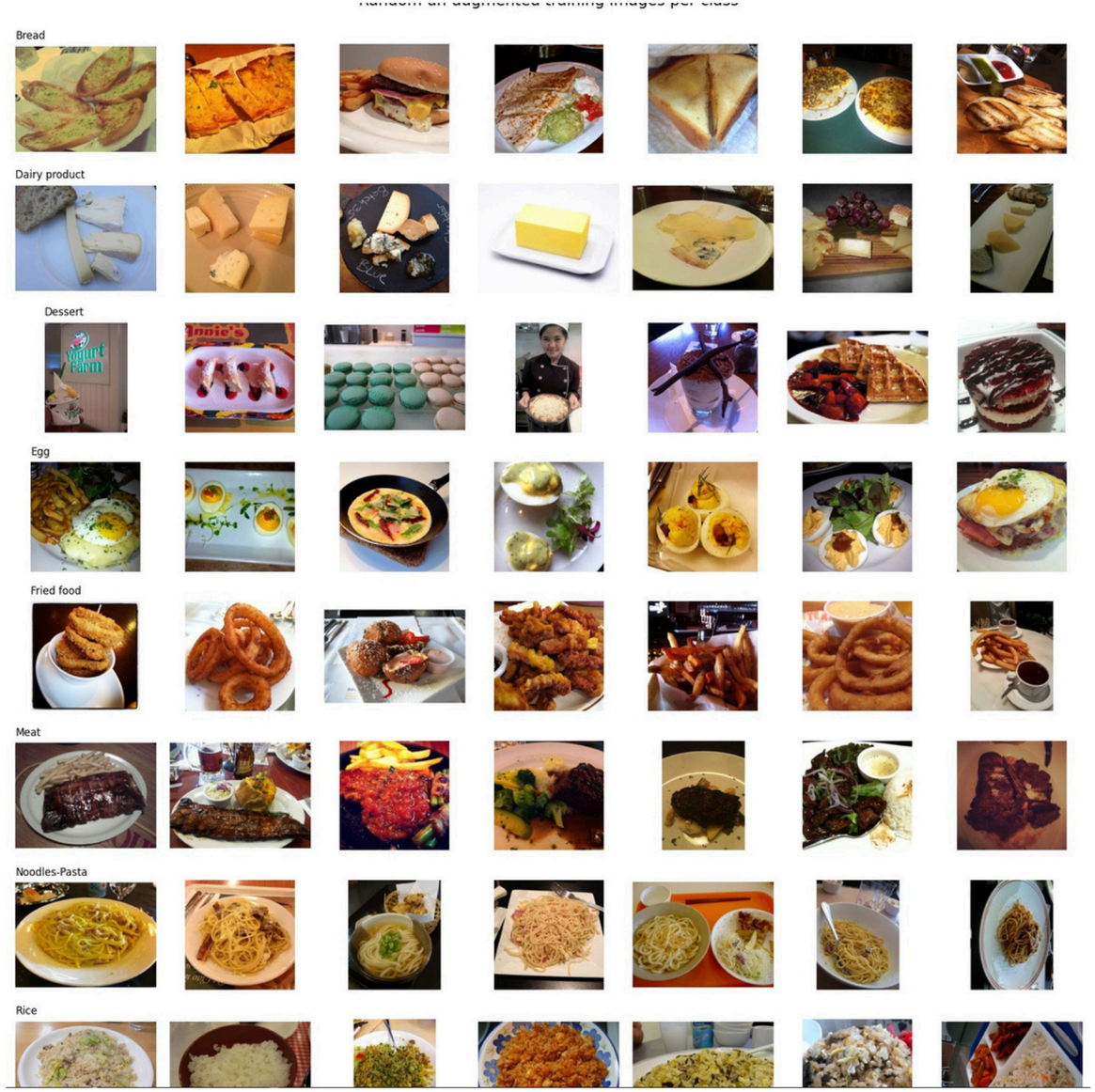


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1. Problem and Data

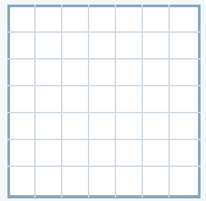
- Problem formulation
- Dataset Description
- Exploratory Data Analysis
- Data Splitting
- Data Preprocessing

Problem Formulation

Single-Label Multiclass Food-11 Classification

INPUT

Input Image



256 × 256 RGB

3 × 256 × 256

→

INPUT REPRESENTATIONS

From scratch	→	**SimpleCNN**
Frozen features	→	**ResNet-18**
Fine-tuned features	→	**ResNet-18**
RGB input	→	**11 classes**

→

MODEL & SOFTMAX

Logits: 11 logits

$$p(k|x) = \frac{exp(z_k)}{\Sigma_k\, exp(z_p)}$$

Softmax

$j = 0, \ldots, 10$

→

OUTPUT

Predicted Class

argmax p(k|x)

$y \in \{0, \ldots, 10\}$

Research Objective: Compare ImageNet transfer learning and backbone fine-tuning on the same cleaned Food-11 data.

Dataset Description: Food-11

EPFL Food-11, Kaggle mirror version 1

16,643
Original images before duplicate removal

11 Classes
Largest / smallest class ratio: 5.30

3 × 256 × 256
RGB, FP32, ImageNet normalization

9,866 / 3,430 / 3,347
Original train / validation / test images

Image and Label Format

Food Photographs — **Class Folders** — **RGB JPEG** — **11 Categories**

Dataset and Annotation

- Food-11: one broad food category per image.
- Evaluation compares SimpleCNN and ResNet-18 transfer learning.

11 Food Categories

Bread, Dairy product, Dessert, Egg, Fried food, Meat, Noodles-Pasta, Rice, Seafood, Soup, Vegetable-Fruit

Key Difficulties & Dataset Limitations

Visual Challenges in Food-11 Recognition

Variation Within a Class

Preparation, viewpoint and lighting change food appearance.

One category can contain very different dishes and backgrounds.

Similarity Between Classes

- Bread, Egg and Fried food can share golden-brown textures.
- Dairy product and Dessert can share creamy textures and plating.

Broad Labels and Mixed Dishes

Each image has one label even when several foods are present.

Labels do not describe ingredients, portions or nutrition.

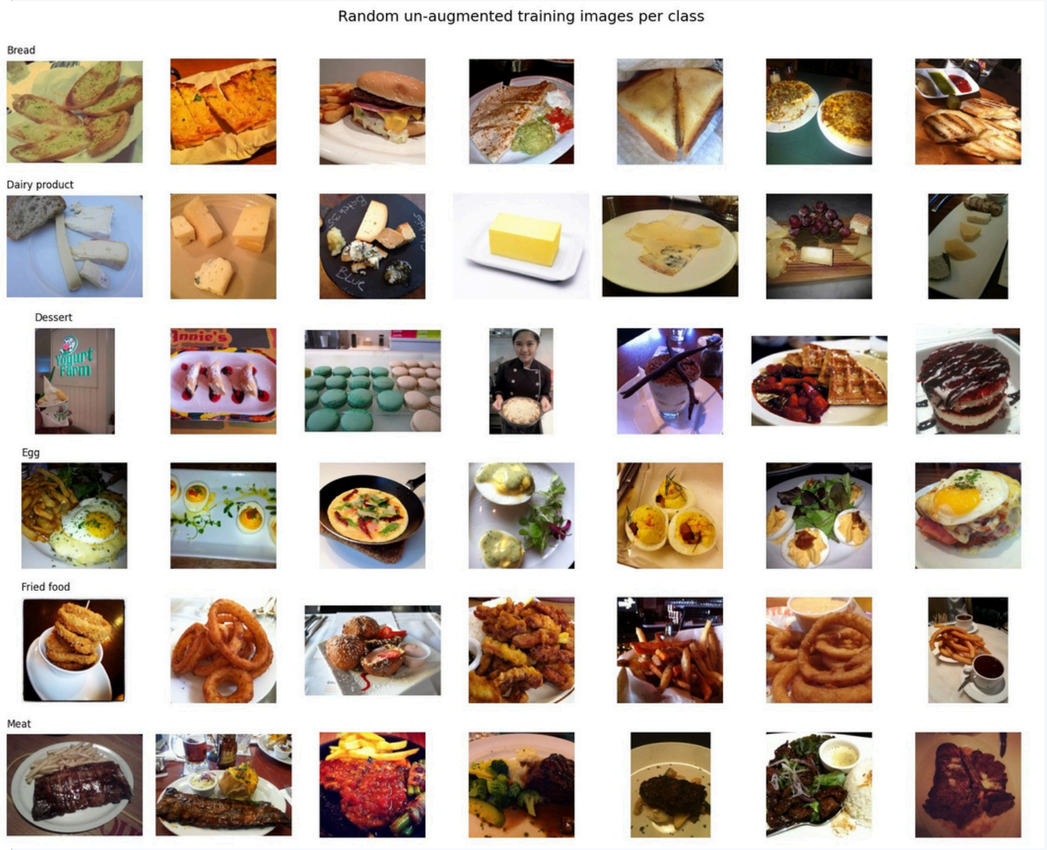
Class Imbalance and Data Quality

- The largest original classes have 5.30 times as many images as Rice.
- Cross-split duplicates require cleaning before evaluation.

EDA: Input and Representative Samples

Random, non-augmented training images from all 11 Food-11 categories

SAMPLE VARIETY WITHIN AND BETWEEN CLASSES





RGB images with variable dimensions. Median: 512 × 512 pixels. Model input: 3 × 256 × 256.

EDA: Class Distribution and Imbalance

Class	Train · count / %	Validation · count / %	Test · count / %
Bread	994 · 10.08%	362 · 10.55%	368 · 10.99%
Dairy product	429 · 4.35%	144 · 4.20%	148 · 4.42%
Dessert	1,500 · 15.20%	500 · 14.58%	500 · 14.94%
Egg	986 · 9.99%	327 · 9.53%	335 · 10.01%
Fried food	848 · 8.60%	326 · 9.50%	287 · 8.57%
Meat	1,325 · 13.43%	449 · 13.09%	432 · 12.91%
Noodles-Pasta	440 · 4.46%	147 · 4.29%	147 · 4.39%
Rice	280 · 2.84%	96 · 2.80%	96 · 2.87%
Seafood	855 · 8.67%	347 · 10.12%	303 · 9.05%
Soup	1,500 · 15.20%	500 · 14.58%	500 · 14.94%
Vegetable-Fruit	709 · 7.19%	232 · 6.76%	231 · 6.90%
TOTAL	**9,866 · 100%**	**3,430 · 100%**	**3,347 · 100%**

Each percentage is calculated within its split.

Counts and shares by split

Train: 9,866 images

Validation: 3,430 images

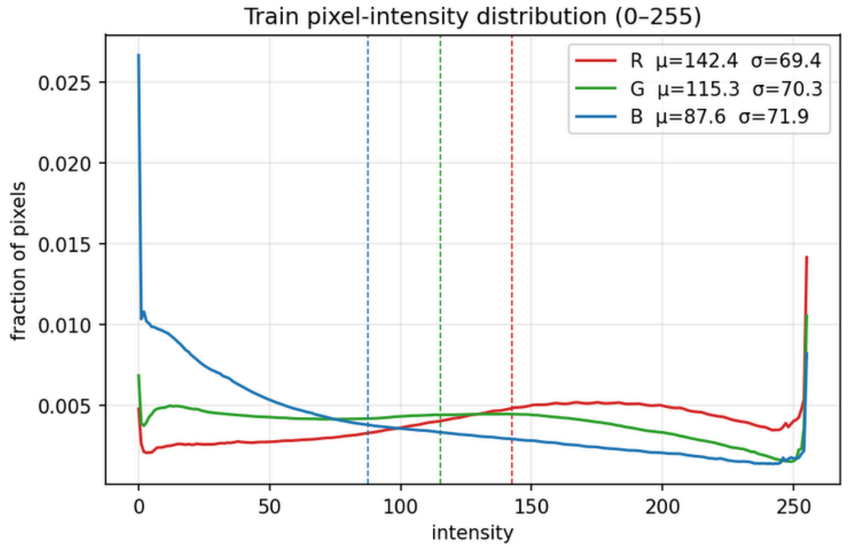
Test: 3,347 images

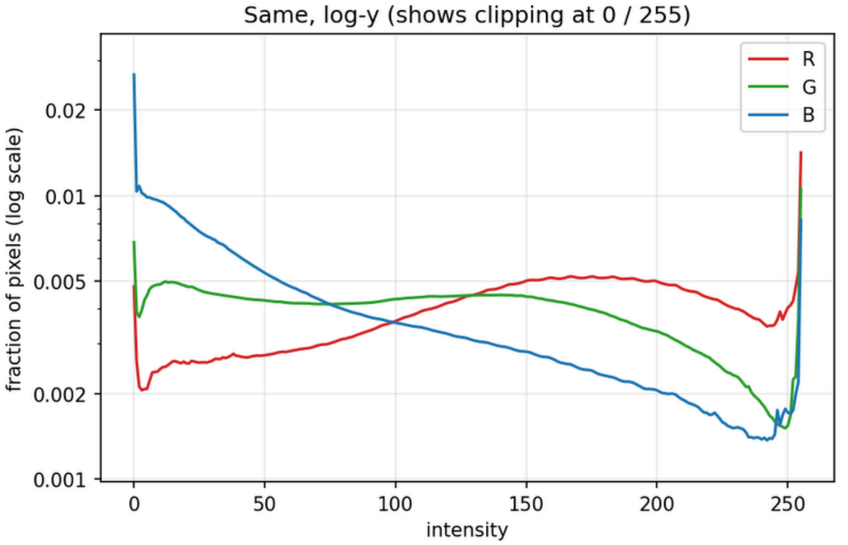
Class shares are similar across splits.
Largest: Dessert and Soup
Smallest: Rice
Largest / smallest ratio: Train 5.36; Val/Test 5.21.

No resampling; unweighted cross-entropy.

EDA: Pixel Intensities

RGB CHANNEL DISTRIBUTIONS AND NORMALIZATION





Original training split only, padding excluded from the analysis

9,866 training images

Channel means on the 0–255 scale
R: 142.4 G: 115.3 B: 87.6

Channel standard deviations on the 0–255 scale
R: 69.4 G: 70.3 B: 71.9

Fixed ImageNet normalization
Mean: (0.485, 0.456, 0.406)
Std: (0.229, 0.224, 0.225)

Data Splitting and Leakage Control

15,980 cleaned images

Food-11



Predefined partitions, no random re-splitting. Total removed: 663 images.

MD5 exact duplicates, pHash distance ≤ 5, plus two manually removed images.

Data Preprocessing

Identical deterministic input processing, with augmentation restricted to training

CLEANED SPLITS, RGB INPUTS AND TRAINING AUGMENTATION

01

CLEANED SPLITS

Training: 9,846 images
Validation: 3,364 images
Test: 2,770 images

Predefined partitions
No random re-splitting

02

RGB RESIZING AND PADDING

Convert each image to RGB

Resize longer side to 256 pixels
Preserve the aspect ratio

Center on a black 256 × 256 canvas

03

AUGMENTATION AND NORMALIZATION

Training-only horizontal flip, p = 0.5

No random augmentation for validation or test

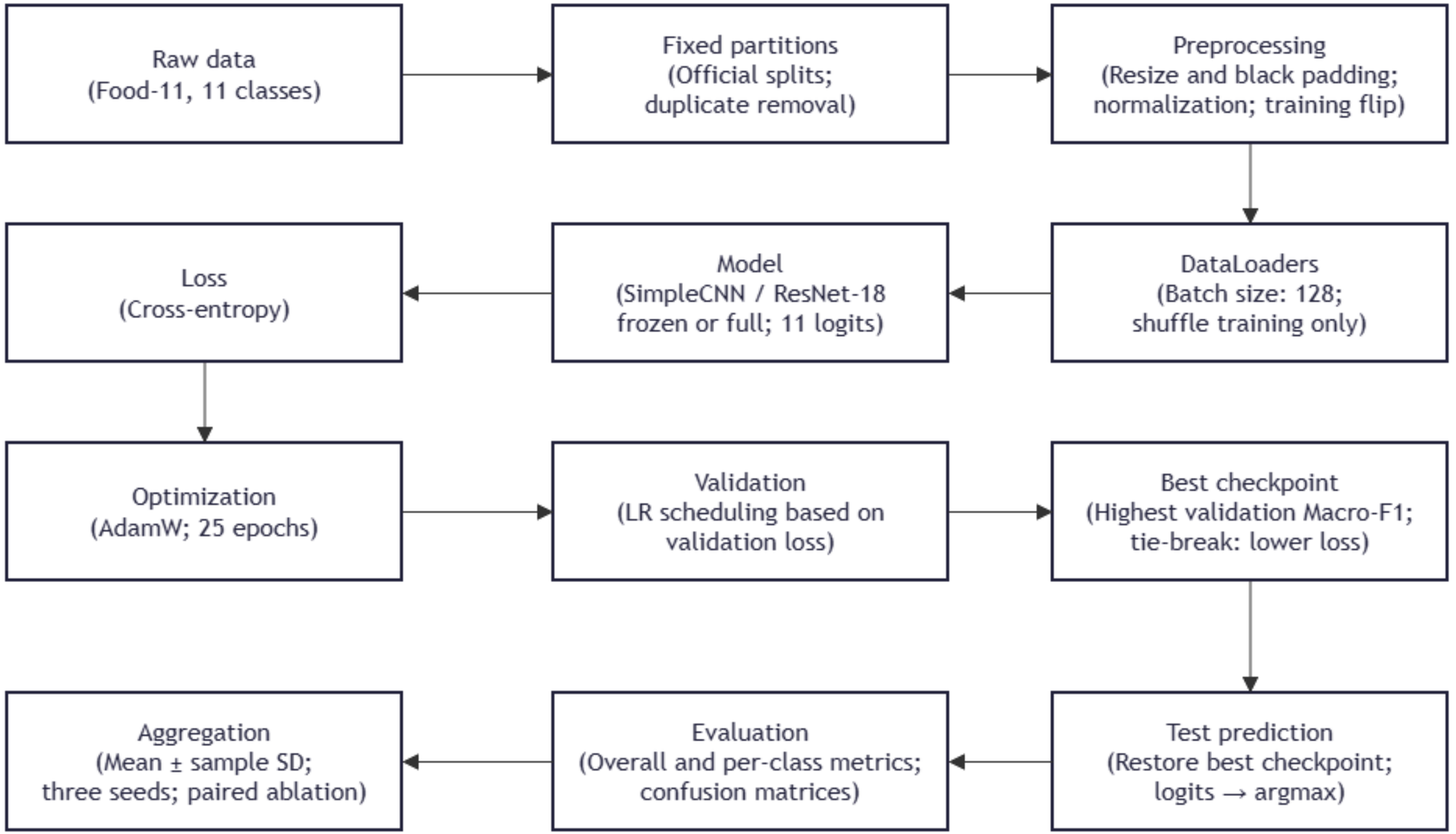
FP32 tensors with fixed ImageNet mean and standard deviation

2. Main Methods

- Pipeline Overview
- SimpleCNN trained from scratch
- SimpleCNN parameter budget
- ResNet-18 architecture and residual learning
- Frozen backbone and full fine-tuning
- Loss, evaluation and checkpoint selection

Pipeline Overview

Shared data and evaluation protocol for three model configurations



SimpleCNN trained from scratch

INPUT REPRESENTATION

RGB input tensor: 3 × 256 × 256

ARCHITECTURE

4 Convolutional blocks

Conv 3x3, BatchNorm, ReLU
max pooling

- Channels: 16, 32, 64, 128
- Global average pooling
- Dropout p = 0.3
- Linear head: 128 to 11 logits

STRENGTHS & LIMITATIONS

STRENGTHS

- Local filters share weights
- Compact pooling-based head
- Low parameter count

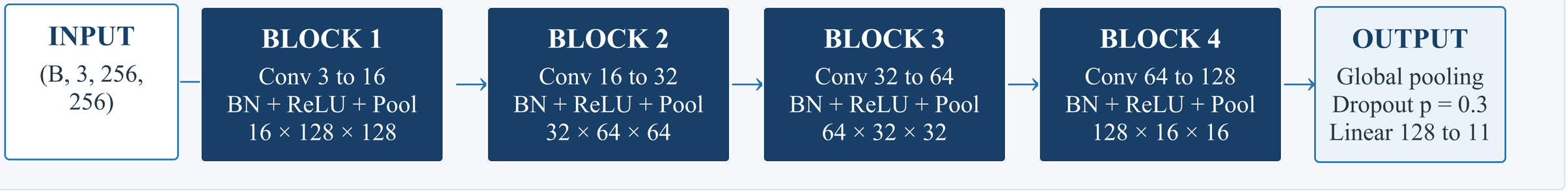
LIMITATIONS

- No pretrained visual features
- Four pooling stages reduce detail
- Performance depends on capacity and training budget

SimpleCNN: Forward Pass

Four convolutional blocks followed by global pooling and an 11-class head

FORWARD PASS ARCHITECTURE



ACTIVATION & REGULARIZATION DESIGN

ReLU introduces nonlinear activations after each convolution.

BatchNorm in all four blocks.
Head dropout p = 0.3 during training only.

LOSS & EVALUATION PROTOCOL

Loss: **nn.CrossEntropyLoss** (internal log-softmax).

Top-1 accuracy and macro-averaged metrics.
Checkpoint: maximum validation Macro-F1.

SimpleCNN: Parameter Budget

COMPONENT	PARAMETERS
Block 1 — Conv + BatchNorm	480
Block 2 — Conv + BatchNorm	4,704
Block 3 — Conv + BatchNorm	18,624
Block 4 — Conv + BatchNorm	74,112
Linear classifier	1,419
Total trainable	**99,339**

FORMULAS

Convolution:

$$C_{out} \times C_{in} \times k^2 + C_{out}$$

BatchNorm:

$$2 \times C_{out}$$

ResNet-18 with ImageNet Pretraining

INPUT REPRESENTATION

RGB input tensor: 3 × 256 × 256

ARCHITECTURE

4 Residual Stages

$$y = (\mathrm{ReLU}(F(x) + x))$$

2 basic blocks per stage
Two 3 × 3 convolutions per block
BatchNorm, ReLU, and shortcut connections

- Stem: 7 × 7 convolution + max pooling
- Channels: 64, 128, 256, 512
- Global average pooling
- Dropout p = 0.3
- Linear head: 512 to 11 logits

STRENGTHS & LIMITATIONS

STRENGTHS

- Pretrained features transfer to Food-11
- Residual connections support gradient flow
- Full fine-tuning adapts features to the task

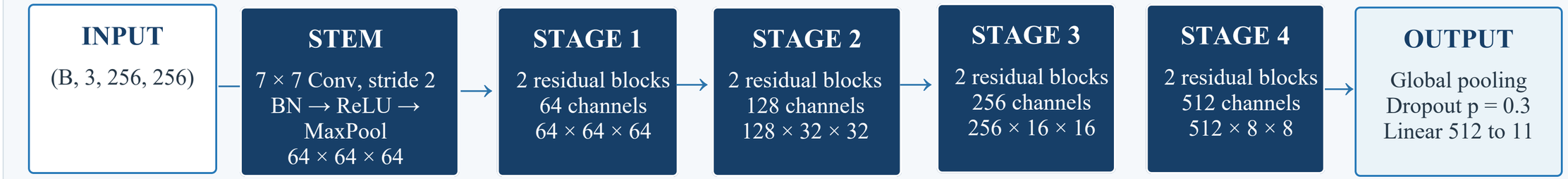
LIMITATIONS

- Higher parameter count than SimpleCNN
- Frozen features have limited task adaptation
- Full fine-tuning increases training cost and may overfit

Resnet18: Forward Pass

A convolutional stem followed by four residual stages, global average pooling, and an 11-class head.

FORWARD PASS ARCHITECTURE



ACTIVATION & REGULARIZATION DESIGN

Head dropout p = 0.3 during training.
Frozen backbone: BatchNorm stays in eval mode.
Full fine-tuning: BatchNorm updates during training.

LOSS & EVALUATION PROTOCOL

Loss: **nn.CrossEntropyLoss** (internal log-softmax).

Top-1 accuracy and macro-averaged metrics.
Checkpoint: maximum validation Macro-F1.

ResNet-18: Parameter Budget

COMPONENT	PARAMETERS
Stem	9,536
Stage 1	147,968
Stage 2	525,568
Stage 3	2,099,712
Stage 4	8,393,728
Classifier	5,643
Total	**11,182,155**

Frozen backbone: 5,643 trainable parameters.
Full fine-tuning: 11,182,155 trainable parameters.
Both variants retain the same total parameter count.

ResNet-18: Frozen Backbone

SETTING	FROZEN CONFIGURATION
Backbone	ImageNet-1K V1
Backbone updates	Frozen
Trainable parameters	5,643
BatchNorm mode	Evaluation
Head dropout	Active; $p = 0.3$
Head learning rate	10^{-3}

Purpose: evaluate transfer of pretrained features with a newly trained classification head.

ResNet-18: Full Fine-tuning

SETTING	FULL FINE-TUNING
Backbone	ImageNet-1K V1
Parameters updated	Backbone + head
Trainable parameters	11,182,155
Backbone learning rate	10^{-4}
Head learning rate	10^{-3}

The frozen and full variants use the same architecture and new 11-class head.

Shared Training Setup

SETTING	CONFIGURATION
Optimizer	AdamW
Weight decay	10^{-2}
Batch size	128
Epoch budget	25; no early stopping
Scheduler	ReduceLROnPlateau
LR factor / patience	0.5 / 3 epochs
Training seed	**42,43,44**
Precision / hardware	**FP32 / Tesla T4**

Scheduler monitors validation loss, minimum LR 10^{-6}. SimpleCNN initial LR: 10^{-3}. Every configuration uses the same cleaned Food-11 partitions and preprocessing.

Loss, Evaluation and Checkpoint Selection

EVALUATION ITEM	RULE / REPORTING
Training loss	Cross-entropy
Primary metric	Test Macro-F1 (11 classes)
Secondary metrics	Top-1 accuracy; macro P/R
Per-class metrics	Precision, recall, F1
Test loss	Cross-entropy
Checkpoint selection	Best validation Macro-F1
Tie-break rule	**Lower validation loss**
Final evaluation	**Restore best model**

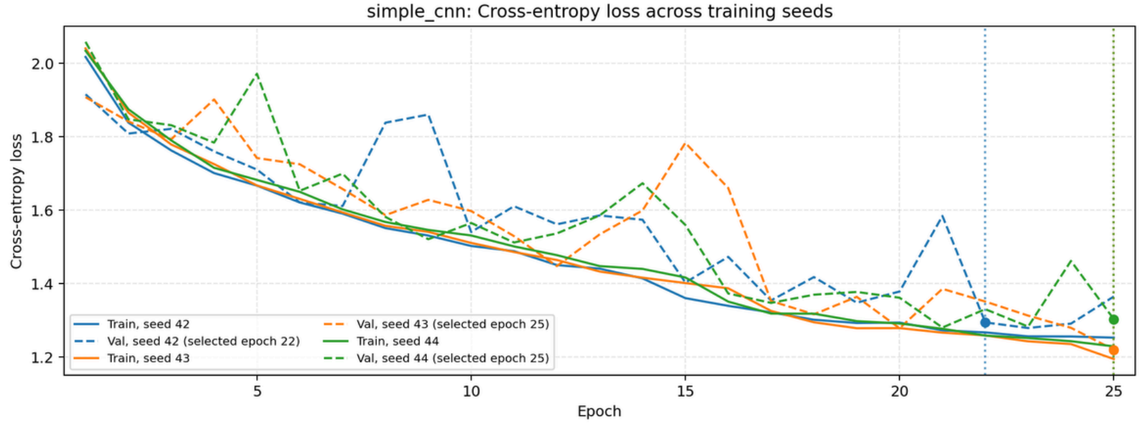
Test results and training times are reported as mean ± sample SD across seeds 42–44. The cleaned test split contains 2,770 images.

3. Implementation Results

- Training Behaviors
- Quantitative Results
- Qualitative Results
- Error Analysis
- Horizontal-Flipping Ablation Study
- Conclusion

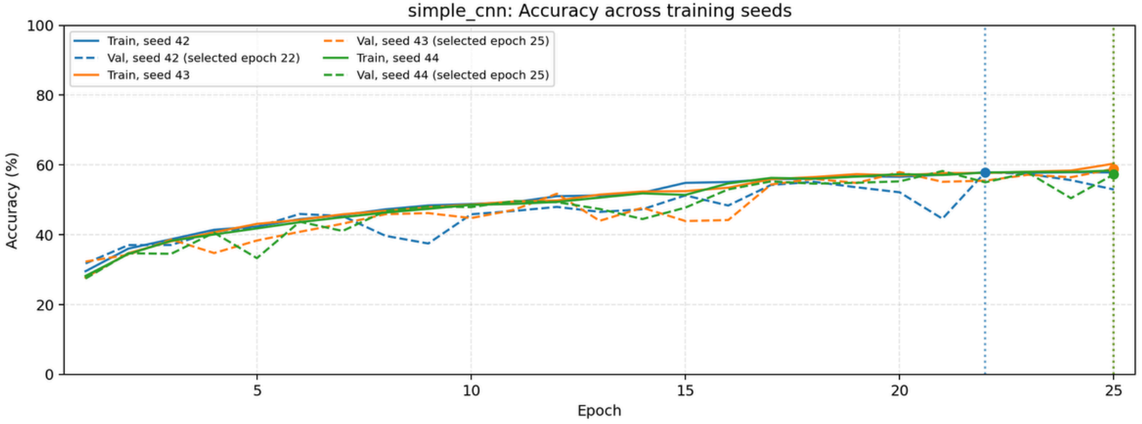
SimpleCNN Training Curves

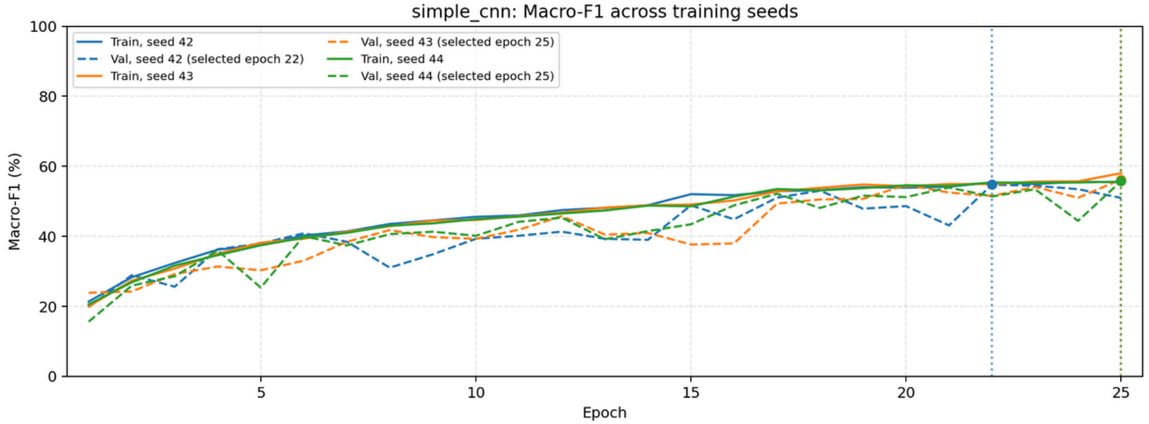
Three seeds • validation-selected checkpoints marked



CHECKPOINTS & INTERPRETATION

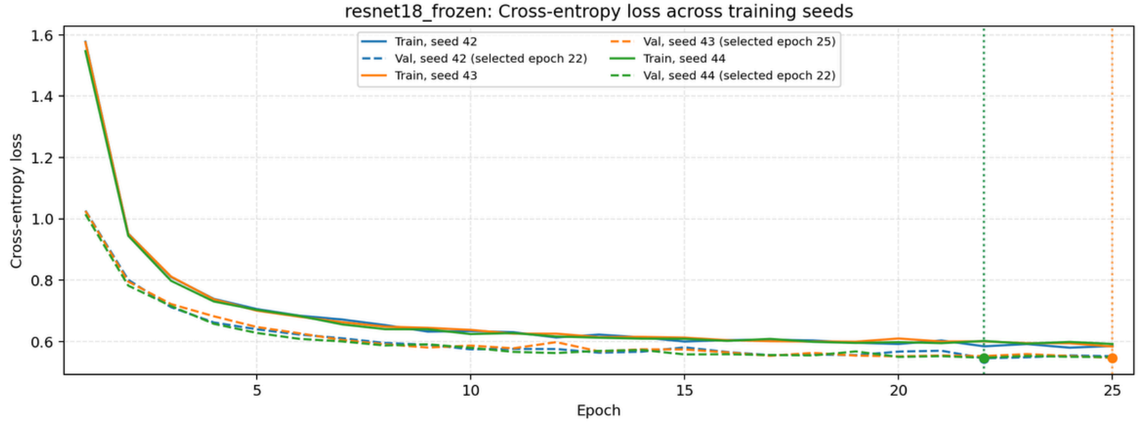
Selected epochs 22 / 25 / 25 (seeds 42–44)
Val Macro-F1 54.74% / 56.22% / 55.74%
Still improving late in the 25-epoch budget • 118.78 ± 4.49 s/epoch.

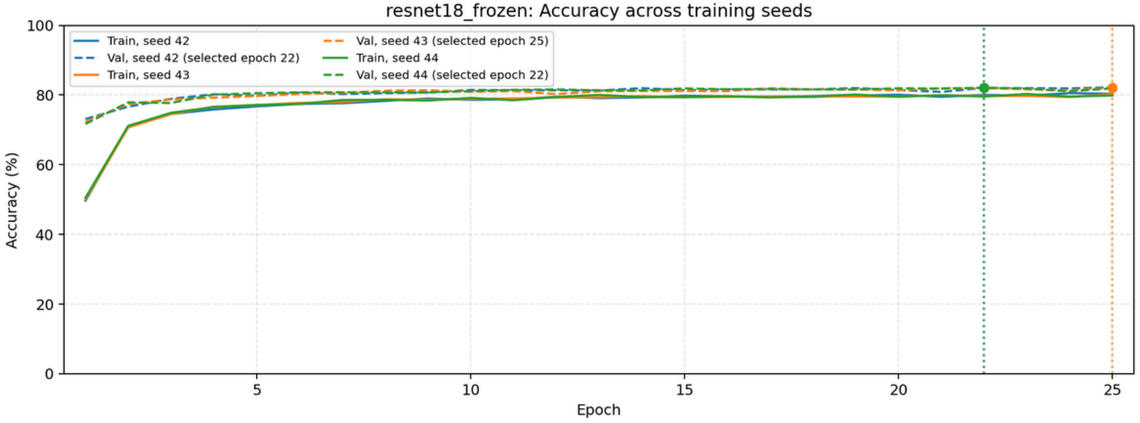


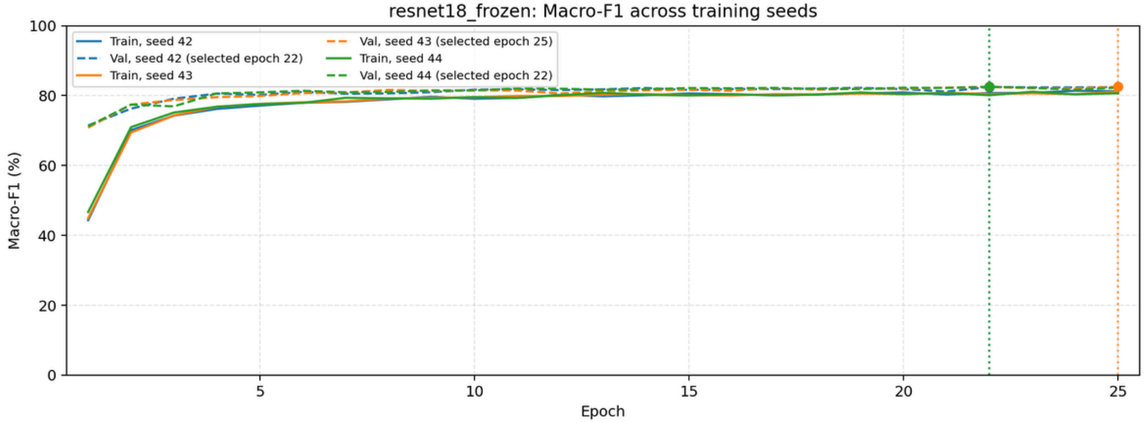


ResNet-18 (Frozen) Training Curves

Three seeds • validation-selected checkpoints marked







CHECKPOINTS & INTERPRETATION

Selected epochs 22 / 25 / 22 (seeds 42–44)
Val Macro-F1 82.47% / 82.54% / 82.54%
Later epochs largely plateau • 118.81 ± 3.89 s/epoch.

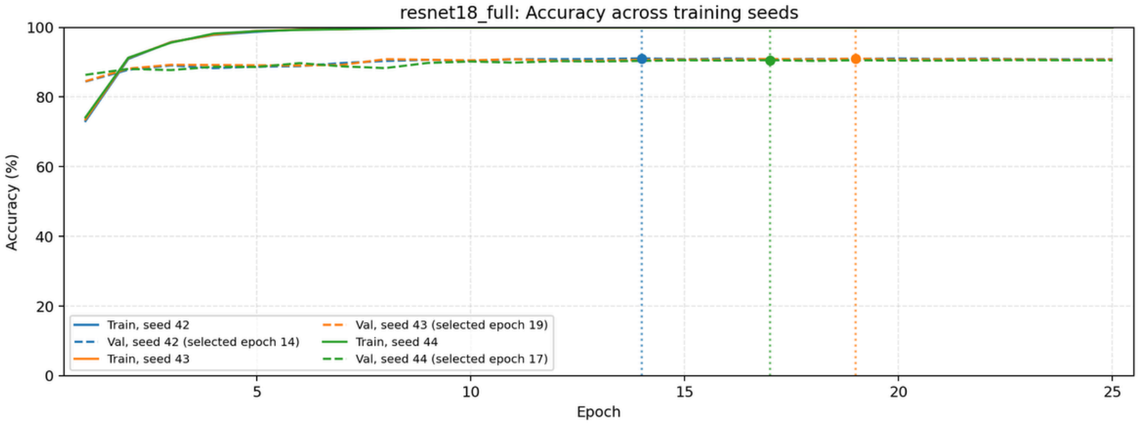
ResNet-18 (Full Fine-Tuning) Curves

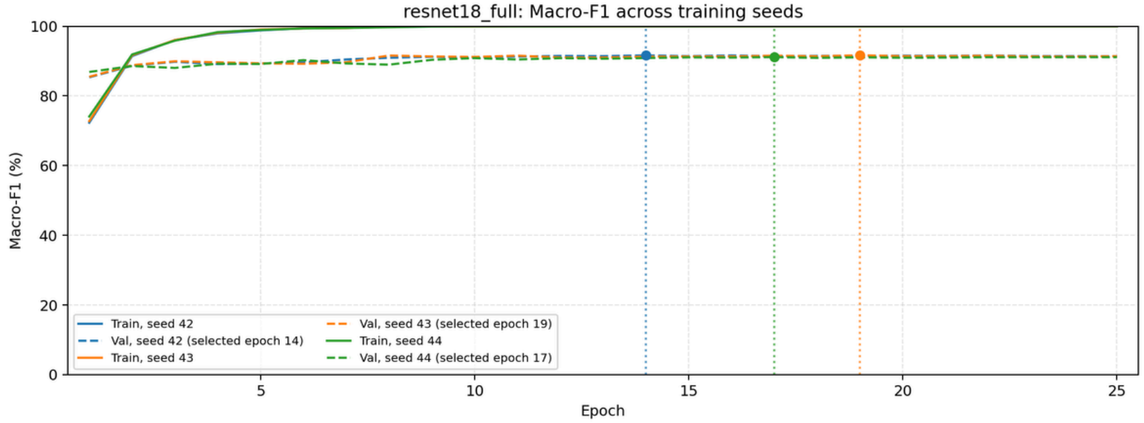
Three seeds • validation-selected checkpoints marked



CHECKPOINTS & INTERPRETATION

Selected epochs 14 / 19 / 17 (seeds 42–44)
Val Macro-F1 91.62% / 91.65% / 91.12%
Late train–validation gap suggests overfitting • 144.78 ± 2.89 s/epoch.





Overall Predictive Results

Predictive performance					
Model	Top-1 Accuracy (%)	Macro-Precision (%)	Macro-Recall (%)	Macro-F1 (%)	Test CE loss
SimpleCNN	60.40 ± 1.02	63.48 ± 2.25	57.23 ± 0.51	57.58 ± 0.84	1.2060 ± 0.0273
ResNet-18 (frozen)	84.62 ± 0.24	85.45 ± 0.10	85.23 ± 0.19	85.26 ± 0.17	0.4938 ± 0.0036
ResNet-18 (full)	$\mathbf{91.37} \pm 0.43$	$\mathbf{91.65} \pm 0.37$	$\mathbf{91.74} \pm 0.40$	$\mathbf{91.67} \pm 0.39$	$\mathbf{0.3197} \pm 0.0148$

Model Size, Training Time & Inference

Model size and computational cost

Model	Total parameters	Trainable parameters	Epochs	Total train time per run (s)	Train time per epoch (s)	Inference (ms/sample)
SimpleCNN	99,339	99,339	25	$2{,}969.5 \pm 112.3$	118.78 ± 4.49	0.8054 ± 0.1036
ResNet-18 (frozen)	11,182,155	5,643	25	$2{,}970.3 \pm 97.2$	118.81 ± 3.89	2.6574 ± 0.4641
ResNet-18 (full)	11,182,155	11,182,155	25	$3{,}619.5 \pm 72.1$	144.78 ± 2.89	2.4819 ± 0.1453

Per-Class Metrics

SimpleCNN

Class	OvR Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Bread	89.64 ± 2.19	48.92 ± 8.70	47.51 ± 11.44	46.87 ± 4.00
Dairy product	95.38 ± 0.07	76.47 ± 3.09	18.82 ± 2.75	30.10 ± 3.44
Dessert	83.25 ± 1.52	48.39 ± 3.52	66.67 ± 6.43	55.83 ± 0.99
Egg	90.29 ± 0.58	61.41 ± 4.42	50.46 ± 3.70	55.22 ± 1.43
Fried food	92.07 ± 1.39	51.22 ± 6.90	64.71 ± 5.56	56.74 ± 2.42
Meat	92.77 ± 0.33	74.98 ± 8.61	59.61 ± 15.22	64.87 ± 7.50
Noodles-Pasta	95.85 ± 0.85	62.09 ± 9.56	61.68 ± 3.49	61.44 ± 3.66
Rice	96.57 ± 0.66	50.42 ± 15.44	39.63 ± 6.70	43.00 ± 2.99
Seafood	94.33 ± 0.32	65.91 ± 10.97	54.52 ± 14.53	57.81 ± 5.87
Soup	92.80 ± 1.58	75.20 ± 9.87	74.57 ± 5.36	74.37 ± 2.88
Vegetable-Fruit	**97.85** ± 0.30	**83.26** ± 3.21	**91.36** ± 0.45	**87.10** ± 1.55

ResNet-18 (Frozen)

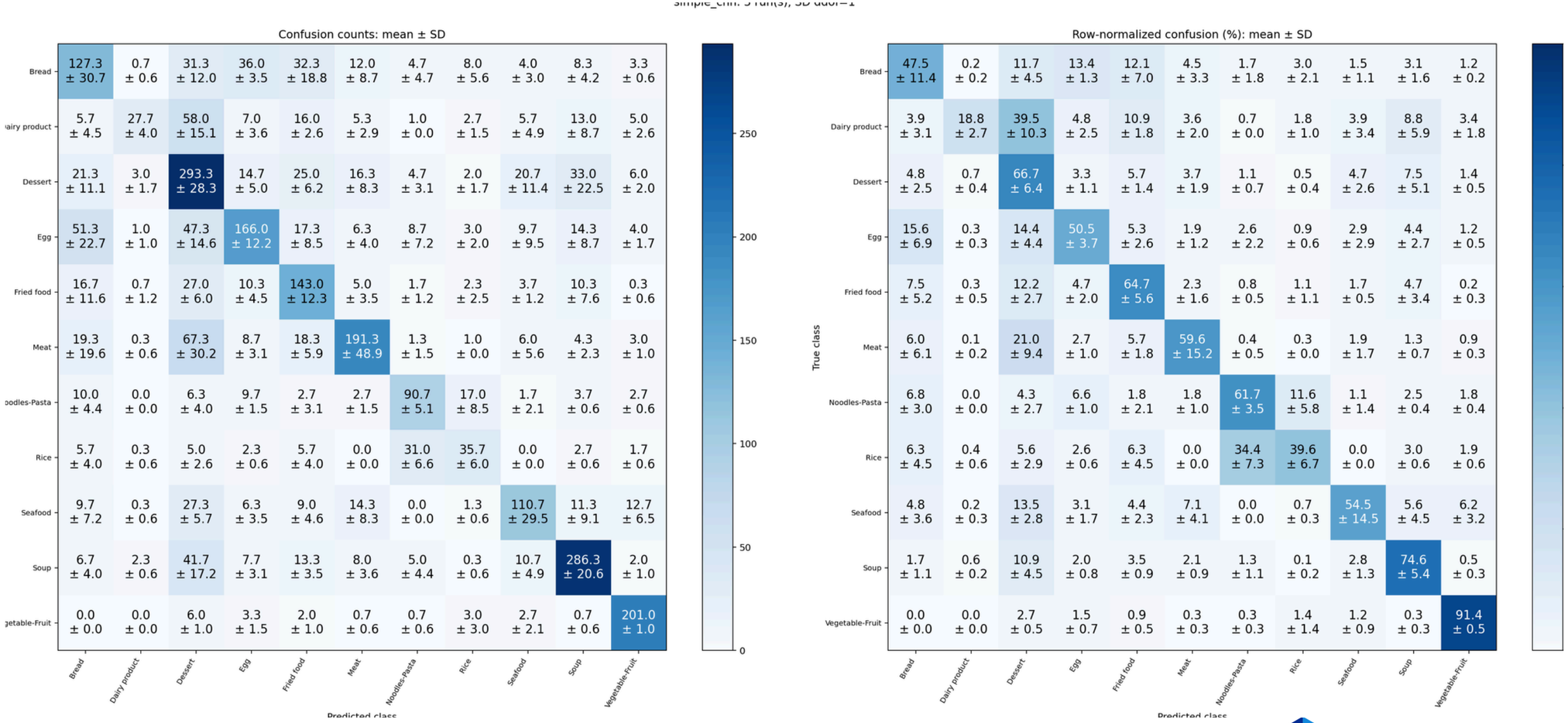
Class	OvR Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Bread	95.28 ± 0.15	75.66 ± 2.39	75.75 ± 2.45	75.65 ± 0.24
Dairy product	97.57 ± 0.09	81.64 ± 2.73	70.07 ± 1.80	75.37 ± 0.41
Dessert	93.39 ± 0.06	80.12 ± 0.93	77.73 ± 2.02	78.89 ± 0.58
Egg	95.09 ± 0.10	79.90 ± 1.12	78.42 ± 2.19	79.13 ± 0.70
Fried food	97.20 ± 0.15	82.07 ± 2.25	83.11 ± 2.23	82.55 ± 0.75
Meat	96.79 ± 0.23	83.56 ± 1.90	90.03 ± 0.62	86.66 ± 0.74
Noodles-Pasta	**99.70** ± 0.02	**96.64** ± 0.01	**97.73** ± 0.39	**97.18** ± 0.20
Rice	99.49 ± 0.04	90.43 ± 0.92	94.44 ± 1.11	92.39 ± 0.55
Seafood	97.45 ± 0.13	82.41 ± 1.77	82.92 ± 0.57	82.66 ± 0.63
Soup	98.33 ± 0.08	92.90 ± 0.82	95.23 ± 0.66	94.04 ± 0.28
Vegetable-Fruit	98.95 ± 0.06	94.59 ± 1.47	92.12 ± 1.84	93.32 ± 0.45

ResNet-18 (Full Fine-Tuning)

Class	OvR Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Bread	97.36 ± 0.23	87.37 ± 1.13	85.07 ± 2.33	86.19 ± 1.30
Dairy product	98.30 ± 0.07	86.59 ± 0.36	80.50 ± 1.42	83.43 ± 0.83
Dessert	96.04 ± 0.12	87.51 ± 0.69	87.58 ± 0.47	87.54 ± 0.33
Egg	97.91 ± 0.25	91.14 ± 2.11	91.29 ± 0.46	91.20 ± 0.96
Fried food	98.23 ± 0.07	88.63 ± 0.66	89.29 ± 1.71	88.95 ± 0.58
Meat	97.79 ± 0.24	89.47 ± 1.29	91.69 ± 0.72	90.57 ± 0.99
Noodles-Pasta	**99.89** ± 0.04	**99.10** ± 0.78	98.87 ± 0.39	**98.98** ± 0.34
Rice	99.72 ± 0.02	93.03 ± 0.56	**98.89** ± 0.00	95.87 ± 0.30
Seafood	98.68 ± 0.08	90.71 ± 0.73	91.30 ± 1.14	91.00 ± 0.54
Soup	99.23 ± 0.08	97.30 ± 0.15	97.14 ± 0.52	97.22 ± 0.28
Vegetable-Fruit	99.59 ± 0.06	97.28 ± 0.45	97.58 ± 0.26	97.43 ± 0.34

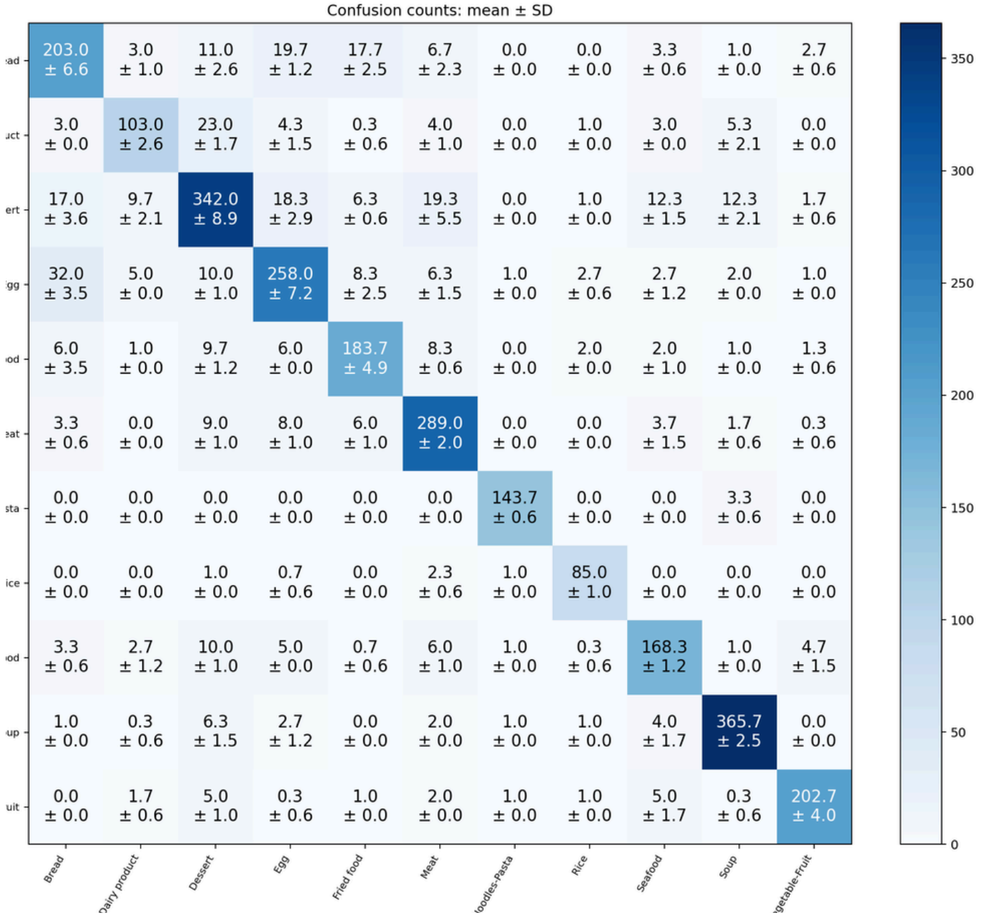
Test Confusion Matrices • SimpleCNN

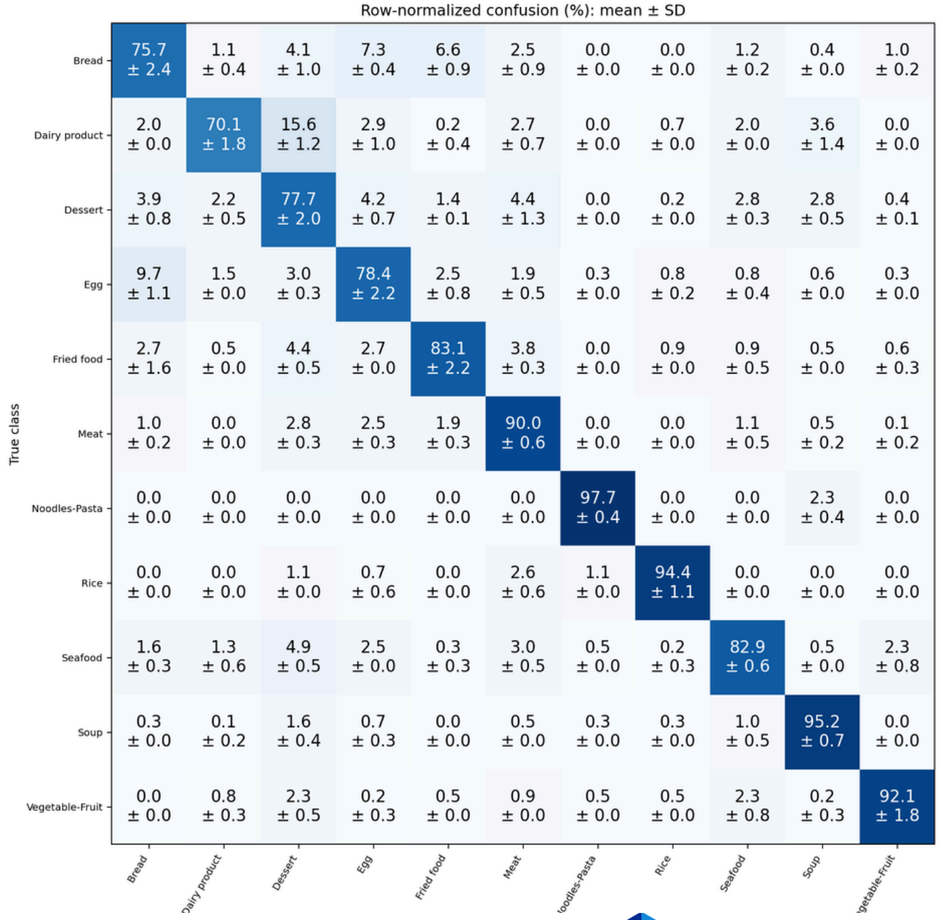
Rows: true labels • columns: predictions • counts and row-normalized rates • mean across seeds 42–44



Test Confusion Matrices • ResNet-18 (frozen)

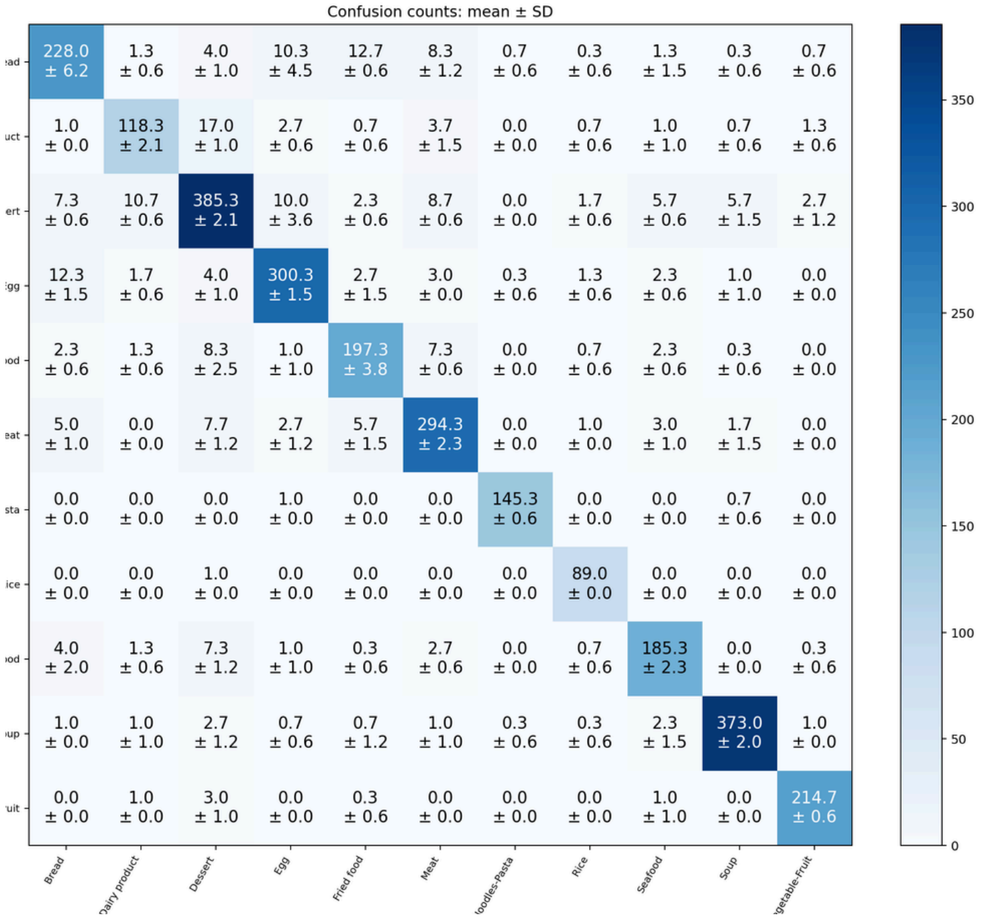
Rows: true labels • columns: predictions • counts and row-normalized rates • mean across seeds 42–44

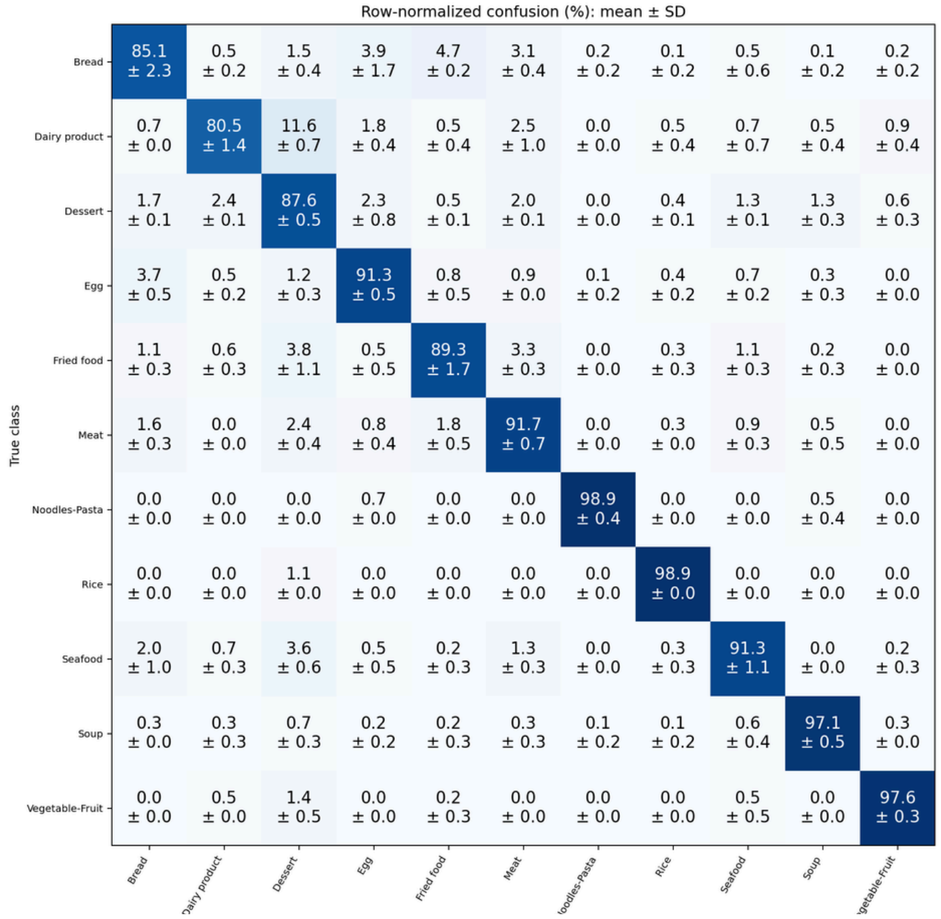




Test Confusion Matrices • ResNet-18 (fully fine-tuned)

Rows: true labels • columns: predictions • counts and row-normalized rates • mean across seeds 42–44





Qualitative Results: Failure Examples

SimpleCNN



Frozen Resnet18

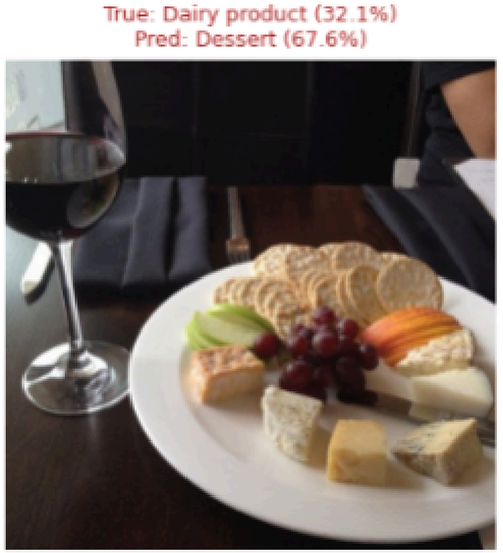


Full Resnet18



Error Analysis: Dairy–Dessert Overlap

Error group	Errors	Main pairs (True → predicted)	Hypothesized cause	Improvement idea
G1. Dairy–Dessert overlap	28 (11.6%)	Dairy product → Dessert (17) Dessert → Dairy product (11)	Shared ingredients, creamy textures, and similar serving styles may blur the boundary between dairy products and desserts. Accompanying foods can further obscure the intended class.	Add training examples of dairy products and visually similar desserts in varied contexts. Review labeling consistency for mixed dairy dishes.









Error Analysis: Bread in Mixed Dishes

G2. Bread in mixed dishes	71 (29.3%)	Bread → Fried food (13) Egg → Bread (12) Bread → Egg (10) Bread → Meat (9) Dessert → Bread (7)	Fillings, toppings, and accompanying foods provide competing class cues. A single image-level label does not explicitly identify the intended main component.	Review the rule for selecting the main food. Compare full-image and main-dish crops to assess the influence of accompanying ingredients.









Error Analysis: Other Classes Predicted as Dessert

G3. Other classes predicted as Dessert	37 (15.3%)	Fried food → Dessert (11) Meat → Dessert (9) Seafood → Dessert (6)	Preparation-based categories can overlap with Dessert. Similar colors, glossy surfaces, and plating styles may also make other foods resemble desserts.	Review labels for sweet fried foods. Add visually similar foods from competing classes and test dish-focused crops.

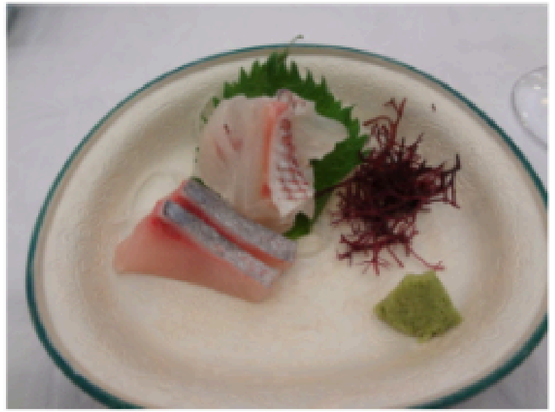
True: Fried food (6.3%)
Pred: Dessert (93.2%)



True: Fried food (11.4%)
Pred: Dessert (86.8%)



True: Seafood (36.5%)
Pred: Dessert (60.5%)



True: Meat (45.4%)
Pred: Dessert (52.3%)



Error Analysis: Desserts Predicted as Other Classes

G4. Desserts predicted as other classes	35 (14.5%)	Dessert → Egg (9) Dessert → Meat (9) Dessert → Soup (6) Dessert → Seafood (5)	Desserts vary substantially in appearance. Color, texture, sauces, and ingredients may resemble other categories. Ambiguous annotations may also contribute to some errors.	Audit ambiguous labels and expand training coverage of visually diverse desserts, especially those resembling other categories.

True: Dessert (6.3%)
Pred: Meat (93.5%)



True: Dessert (0.3%)
Pred: Egg (69.2%)



True: Dessert (33.6%)
Pred: Soup (34.7%)



True: Dessert (32.4%)
Pred: Egg (67.3%)



Error Analysis: Ingredient and Preparation Overlap

G5. Ingredient and preparation overlap	35 (14.5%)	Fried food → Meat (7) Meat → Fried food (4) Meat → Egg (4)	Ingredient-based and preparation-based labels can describe the same dish. Coatings may hide ingredients, while mixed dishes contain cues associated with multiple classes.	Review category precedence for foods satisfying multiple labels. Add consistently labeled examples of coated foods and mixed dishes.









Horizontal-Flip Ablation • Paired Across Three Seeds

Model	Top-1 Accuracy (%)	Macro-Precision (%)	Macro-Recall (%)	Macro-F1 (%)	Test CE loss
SimpleCNN	60.01 ± 1.47	62.41 ± 0.91	56.79 ± 1.43	57.09 ± 1.81	1.2211 ± 0.0196
ResNet-18 (frozen)	84.33 ± 0.06	85.50 ± 0.04	84.52 ± 0.11	84.85 ± 0.11	0.4988 ± 0.0022
ResNet-18 (full)	$\mathbf{90.76} \pm 0.07$	$\mathbf{91.12} \pm 0.14$	$\mathbf{91.18} \pm 0.04$	$\mathbf{91.12} \pm 0.05$	$\mathbf{0.3423} \pm 0.0087$

Model	Δ Top-1 Accuracy (%)	Δ Macro-Precision (%)	Δ Macro-Recall (%)	Δ Macro-F1 (%)	Δ Test CE loss
SimpleCNN	$+0.39 \pm 0.50$	$+1.07 \pm 2.38$	$+0.44 \pm 1.12$	$+0.49 \pm 1.30$	-0.0151 ± 0.0156
ResNet-18 (frozen)	$+0.29 \pm 0.19$	-0.05 ± 0.06	$+0.71 \pm 0.22$	$+0.41 \pm 0.16$	-0.0051 ± 0.0019
ResNet-18 (full)	$+0.61 \pm 0.38$	$+0.53 \pm 0.41$	$+0.56 \pm 0.42$	$+0.55 \pm 0.42$	-0.0227 ± 0.0106

Conclusion

Main Comparision

91.67 ± 0.39% Macro-F1

ResNet-18 full fine-tuning

- **Frozen ResNet-18:** 85.26 ± 0.17%
- **SimpleCNN:** 57.58 ± 0.84%
- Both pretrained configurations outperform SimpleCNN under the evaluated protocol.

Augmentation Ablation Study

- **SimpleCNN:** +0.49 ± 1.30 ΔMacro-F1
- **Frozen ResNet-18:** +0.41 ± 0.16 ΔMacro-F1
- **Full fine-tuning:** +0.55 ± 0.42 ΔMacro-F1
- **Hypothesis 2 supported for ResNet-18:** Gains occur across all three seeds.
- **Inconclusive for SimpleCNN:** The effect varies in direction across seeds.

Practical Trade-offs

- **Full fine-tuning:** Best predictive performance, with higher training cost.
- **Frozen backbone:** Strong performance while training only 5,643 parameters.
- **SimpleCNN:** Compact and fast at inference, but substantially lower Macro-F1.

Fully fine-tuned ResNet-18 with horizontal flipping achieves the best mean test Macro-F1 under the evaluated protocol.